Yeh-Ching Chung, Professor
School of Data Science
Chinese University of Hong Kong in Shenzhen
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Research Interests

- Cloud-Fog Computing: Virtualization Techniques (CPUs, GPUs, Memory, Storage, and Network), Cloud-RAN Design for 5G/IoT Networks
- Big Data: Applications (Agriculture and Large-Scale Software System Optimizations), Graph Computing for Association Rule Mining, and Deep learning for Medical Image Processing
- Embedded Systems: Tool-Chain for Heterogeneous System Architecture (Compilers. Emulators, and Debuggers), Virtualization for Heterogeneous System Architecture, Embedded Virtual Platform
- Parallel and Distributed Systems: Parallelizing Compilers, Parallel I/O and File Systems, P2P, HLA, and Grid Computing

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Education

- Ph.D. in Computer and Information Science (1992), Syracuse University
- M.S. in Computer Science (1988), Syracuse University
- B.S. in Computer Science (1983), Chung-Yuan Christian University

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Experience

- Professor – School of Data Science, Chinese University of Hong Kong in Shenzhen, Shenzhen, Guangdong 518172, China (Since July 2020)
- Professor – School of Science and Engineering, Chinese University of Hong Kong in Shenzhen, Shenzhen, Guangdong 518172, China (September 2018 – June 2020)
- Deputy Director – Cloud Computing and Disaster Recovery Laboratory, Research Institute of Tsinghua University in Shenzhen, Shenzhen, Guangdong 518057, China (October 2015 – August 2018)
- Professor – Department of Computer Science, National Tsing Hua University, Hsinchu, Taiwan 30013, ROC (August 2002 – January 2016)
- Deputy Director – Library, National Tsing Hua University, Hsinchu, Taiwan 30013, ROC (December 2003 – July 2012)
- Deputy Director – Computer and Communication Research Center, National Tsing Hua University, Hsinchu, Taiwan 30013, ROC (August 2008 – July 2015)
- Director General – Taiwan Association of Cloud Computing (March 2007 – February 2011)

- Chairman – Department of Information Engineering and Computer Science, Feng Chia University, Taichung, Taiwan 407, ROC (August 1998 – July 2001)
- Professor – Department of Information Engineering and Computer Science, Feng Chia University, Taichung, Taiwan 407, ROC (July 1999 – July 2002)
- Associate Professor – Department of Information Engineering and Computer Science, Feng Chia University, Taichung, Taiwan 407, ROC (August 1992 – June 1999)
- Director – Networking Division, Computer Center, Feng Chia University, Taichung, Taiwan 407, ROC (January 1994 – July 1997)
- Director – Education Division, Computer Center, Feng Chia University, Taichung, Taiwan 407, ROC (August 1993 – December 1993)

Professional Societies Membership

- IEEE Senior Member, IEEE Computer Society
- ACM, ACM SIGBED
- Taiwan Association Cloud Computing (TACC)

Awards

- 2024 ACM SCC conference - The Outstanding Paper Award. Paper title “Isolate and Detect the Untrusted Driver with a Virtual Box"
- 2024 Outstanding Alibaba Innovative Research Program – Project title “A Large-Scale Graph Pattern Matching Query System and Optimizations”
- 2021 CUHK-SZ Ethics of Teachers Construction Award
- 2017 IEEE SC2 conference - The Outstanding Paper Award. Paper title “Application-Aware Traffic Redirection: A Mobile Edge Computing Implementation Toward Future 5G Networks"
- 2017 IEEE ICASI conference - The First Prize Paper Award. Paper title “File Placement Mechanisms for Improving Write Throughputs of Cloud Storage Services Based on Ceph and HDFS"
- 2016 IEEE SC2 conference - The Best Poster Award. Poster title "Byzantine Fault Tolerant Optimization in Federated Cloud Computing"
- 2016 ACM VEE conference - The Outstanding Paper Award. Paper title “Building A KVM-based Hypervisor for A Heterogeneous System Architecture Compliant System”
- 2015 China Thousand Talents Plan Awardee (innovative long-term)
- 2014 NTHU Distinguished Industry-Academy Cooperation Award for excellent contribution to the Big Data
- 2014 ACM/IEEE Supercomputing conference (SC'14) - Student Cluster Competition Highest LINKPACK Award

- 2014 NCHC Student Cluster Competition Champion
- 2013 NCHC Student Cluster Competition Champion
- 2012 NTHU Distinguished Industry-Academy Cooperation Award for excellent contribution to the Cloud Computing
- 2012 NCHC Student Cluster Competition Champion
- 2011 IEEE ICPADS conference - The Best Paper Award. Paper title "PQEMU: A Parallel System Emulator Based on QEMU"
- 2011 ACM/IEEE Supercomputing conference (SC'11) - Student Cluster Competition Overall Winner (Champion)
- 2011 NCHC Student Cluster Competition Champion
- 2010 ACM/IEEE Supercomputing Conference (SC'10) - Student Cluster Competition overall winner (Champion)
- 2010 Embedded System – Hardware/Software Co-Design Contest Champion, Ministry of Education of ROC
- 2008 Chung-Shang Institute of Science and Technology Distinguished Academy Cooperation Award for excellent contribution to the High Level Architecture (HLA) System Design

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Grants (Since 2003)

- The Key Techniques of Workload Generator on Cloud Environments, Huawei，RMB1,190,000, 2022.11-2024.04.
- A Large-Scale Graph Matching Query System and Optimizations, Ali Damou Research Institute, RMB 500,000, 2022.05-2023.04.
- An Agile and Customizable Intelligent Visual Processor and its Applications, Key Area Research and Development Program of Guangdong Province, No. 2019B010155002, RMB 20,000,000, 2020.01-2024.01. (with C.W. Chen et al.)
- A Generic Graph Computing Processor Technique and System, Ministry of Science and Technology, RMB 40,610,000, 2018.05-2020.05. (with X.F. Liao et al.)
- A High Reliable Communication Theory and Technology Under Ocean Environment, NSFC Guangdong joint fund, RMB 2,700,000, 2018.01-2022.01. (with K.Q. Li et al.)
- A Fusion of SQL and NoSQL Databases for Big Data Applications, Guangdong Province, RMB 1,000,000, 2016.10-2018.10.
- A High-Performance Big Data Analytical Platform Based on Graph Computing, Shenzhen City Brach Committee, RMB 1,000,000, 2016.01-2018.01. (with W.G. Chen)
- Smart City, Ministry of Science and Technology, NT$5,000,000, 2014.08-2017.08. (with C.T. King et al.)
- 5G/ToT Cloud Platform, ITRI, NT$2,000,000, 2014.10-2016.04.
- High-Performance Big Data Appliance, National Science Council, NT$8,000,000, 2014.02-2015.02.

- A Hybrid Just-in-Time Compilation for Android System, National Science Council, NT$800,000, 2013.08-2014.08.
- The Automatic Workflow Deployment on Cloud Platform, III, NT$800,000, 2013.04-2014.04.
- Virtualization Techniques for ARM Processors, National Science Council, NT$26,000,000, 2012.12-2016.12.
- Cloud Platforms for HPC Applications, Quanta Computer Corporation, NT$7,900,000, 2012.08-2013.07
- Distributed File System, National Science Council, NT$761,000, 2012.08-2013.07
- DCloud - A Distributed Cloud Middleware, Delta Electronic Corporation, NT$36,000,000, 2011.08-2014.07.
- Development of HLA-based Simulation System using Distributed Computing Techniques, National Science Council, NT$797,000, 2011.01-2011.12.
- UniCloud: A Distributed Cloud Computing Middleware, National Science Council, NT$1,349,000, 2010.08-2011.07.
- UniCloud-Subproject C1: Storage Virtualization, National Science Council, NT$945,000, 2010.08-2011.07.
- A Volunteer Computing System for Computer Games and its Applications, National Science Council, NT$2,528,000, 2010.08-2011.07. (with I.C. Wu et al.)
- The Design of Desktop Grids in the Volunteer Computing System, National Science Council, NT$2,206,000, 2010.08-2013.07. (with I.C. Wu et al.)
- Ontology on Monte Carlo Tree Search and Applications, National Science Council, NT$7,181,000, 2010.01-2012.12. (with C.S. Lee et al.)
- A P2P HLA RTI Design, National Science Council, NT$851,000, 2009.01-2009.12.
- Home-Grid: A Pervasive P2P Based Personal Health-Care Grid System, National Science Council, NT$20,171,000, 2008.11-2011.10.
- NTHU-MTK Embedded System Lab, MediaTek Corporation, NT$60,000,000, 2008.08-2013.07. (with J.M. Shyu et al.)
- ESL Based Developing and Verification Environment for Embedded Multicore Systems, National Science Council, NT$3,226,000, 2008.08-2010.07. (with J.M. Shyu et al.)
- Pisces: A P2P Meta-Grid System, National Science Council, NT$3,260,000, 2007.08-2010.07.
- Virtual Battlefield, Chung-Shang Institute of Science and Technology, NT$4,342,000, 2007.01-2008.12.
- A Wireless Sensor Network for Landslides Monitoring, National Science Council, NT$12,884,000, 2006.11-2009.10. (with Pai Chou et al.)
- MultiFed – A Scalable HLA RTI System, Chung-Shang Institute of Science and Technology, NT$4,220,000, 2006.01-2008.12.
- Medicare Grid – A Grid Based E-health System, National Science Council, NT$20,802,000, 2005.11-2008.10.
- Design a Fat-Tree Based Cluster System, National Science Council, NT$1,221,000, 2005.08-2007.07.

- An Intelligent Sensor Network and its Applications, Ministry of Economic Affair, NT$69,000,000, 2005.06-2008.05. (with C.T. King et al.)
- StarFish – An Ultralow power DSP, Ministry of Economic Affair, NT$55,000,000, 2005.06-2008.05. (with Y.L. Lin et al.)
- Taiwan UniGrid – A University Grid In Taiwan, National Center for High-Performance Computing), NT$4,040,000, 2005.01-2006.12.
- A General-Purpose Sensor Platform Design, National Science Council, NT$4,821,100, 2004.08-2007.07.
- Overlay Network Technology and Applications, National Science Council, NT$27,042,400, 2004.04-2008.03. (with C.T. King et al.)
- A Data Grid System for Data Streaming Applications Based on Microsoft Windows, Microsoft, NT$1,800,000, 2004.01-2005.01
- HPC Technology and Applications Based on Microsoft Windows, Microsoft, NT$2,400,000, 2003.01-2004.01.

Journal Editorship

- Associate Editor, ACM Computing Surveys, Since 2014
- Associate Editor, International Journal of Grid and High-Performance Computing (IJGHPC), Since 2009
- Associate Editor, International Journal of Big Data Intelligence (IJBDI), 2013-2015
- Advisory Committee Member, DataCom conference, since 2019
- Associate Editor, Journal of Chinese Institute of Engineers (JCIE), 2012-2013

Patents

- **Yeh-Ching Chung** and Yuan Yu, "Terminal, Storage Medium, and Database Synchronization Method Thereof," China Patent, No. ZL 2020 1 0933547.2, November 07, 2023.
- **Yeh-Ching Chung** and Yuan Yu, "Terminal, Storage Medium, and Database Synchronization Method Thereof," US Patent, No. US11604805 B2, March 14, 2023.
- **Yeh-Ching Chung**, Lidong Zhang, and Yongwei Wu, "Hybrid File System Architecture, File Storage, Dynamic Migration, and Application Thereof," US Patent, No. US10810169 B1, Oct.20, 2020.
- **Yeh-Ching Chung**, Rui Zhang, and Wenguang Chen, "Method for Computing Support of Itemset Candidate Based on Graph Structure Data and Application Thereof," US Patent, No. US10776372 B2, Sept. 15, 2020.
- Ying-Ti Liao, Jiazheng Zhou, and **Yeh-Ching Chung**, "Synchronization System for Transforming Database and Method Thereof," China Patent, No. 201410776112, October 25, 2018.

- Hung-Chang Hsiao, Chi-Tsun Liao, Chia-Ping Tsai, and **Yeh-Ching Chung**, "System and method for recovering system state consistency to any point-in-time in distributed database," Taiwan patent. No. I626547, Taiwan, June 11, 2018.
- Ying-Ti Liao, Jiazheng Zhou, and **Yeh-Ching Chung**, "Synchronization System for Transforming Database and Method Thereof," US patent, No. US9779154B2, October 3, 2017.
- Hung-Chang Hsiao, Chi-Tsun Liao, Chia-Ping Tsai, and **Yeh-Ching Chung**, "System and method for recovering system state consistency to any point-in-time in distributed database," US patent, No. US9372725B2, June 21, 2016.
- Ying-Ti Liao, Jiazheng Zhou, and **Yeh-Ching Chung**, "Synchronization System for Transforming Database and Method Thereof," Taiwan patent, No. I539309, June 21, 2016.
- **Yeh-Ching Chung** and Jiun-Hung Ding, "Multicore Processing System and its Scheduling Method," Taiwan patent, No. I503742, October 11, 2015.
- Wei-Chih Sun and **Yeh-Ching Chung**, "Cloud Management Systems and Methods for Executing Applications on Android Systems," Taiwan patent, No. I49918, September 9, 2015.

Book Chapters

1. Kuo-Chan Huang, Po-Chi Shih, and **Yeh-Ching Chung,** "Effective Resource Allocation and Job Scheduling Mechanisms for Load Sharing in a Computational Grid", *Handbook of Research on Grid Technologies and Utility Computing: Concepts for Managing Large-Scale Applications*, Edited by Emmanuel Udoh and Frank Zhigang Wang, pp. 31-40, Information Science Publishing, May 31, 2009, (ISBN: 978-1605661841).

2. Kuo-Chan Huang, Po-Chi Shih, and **Yeh-Ching Chung**, "QoS-based Job Scheduling and Resource Management Strategies for Grid Computing," *Quantitative Quality of Service for Grid Computing: Applications for Heterogeneity, Large-Scale Distribution, and Dynamic Environments*, Edited by Lizhe Wang, Jinjun Chen, and Wei Jie, pp. 110-127, Information Science Publishing, May 5, 2009 (ISBN: 978-1605663708).

3. William C. Chu, Chih-Hong Chang, Chih-Wei Lu, H. Yang, H.C. Jiau, **Yeh-Ching Chung**, and B. Qiao, "Enhanceing Software Maintainability by Unifying and Integrating Standards," *Advances in Software Maintenance Management: Technologies and Solutions*, pp. 114-150, Idea Group Publishing, 2002.

4. Chih-Wei Lu, William C. Chu, Chih-Hong Chang, **Yeh-Ching Chung**, X. Liu, and H. Yang, "Reverse Engineering," *Handbook of Software Engineering and Knowledge Engineering*, Vol. 2, pp. 447-466, 2002.

5. William C. Chu, Chih-Wei Lu, Chih-Hong Chang, and **Yeh-Ching Chung**, "Pattern-Based Software Reengineering," *Handbook of Software Engineering and Knowledge Engineering*, Vol. 1, pp.767-786, 2001.

Journal Papers

J1. Yingying Qu, Jiangtao Cui, Haotian Chi, Yonggang Li, **Yeh-Ching Chung**, and Shunrong Jiang, "Fortifying smart contracts through rich heterogeneous graph semantics and dual attention guards," *Engineering Applications of Artificial Intelligence* (EAAI), Vol. 182, Part 4, October 2026, 116074. (SCI, EI, CCF C)

J2. Lele Li, Wei-Chung Hsu, **Yeh-Ching Chung**, and Zhibin Yu, "Improving Resource Efficiency of Performance-optimized In-memory Big Data Analytics with a Hybrid Static Dynamic Approach," *IEEE Transactions on Computers* (TC), Vol. 75, No. 9, September 2026, pp. 3271-3283. (SCI, EI, CCF A)

J3. Shihao Hong and **Yeh-Ching Chung**, "CRPIM: An Efficient Compute-Reuse Scheme for ReRAM-based Processing-in-Memory DNN Accelerators," *Journal of System Architecture* (JSA), 153, August 2024, 103192. (SCI, EI, CCF B)

J4. YongGang Li, Yu Bao, and **Yeh-Ching Chung**, "Randomize the Running Function When It Is Disclosed," *IEEE Transactions on Computers* (TC), Vol. 73, No. 6, 2024, pp. 1516-1530. (SCI, EI, CCF A)

J5. YongGang Li, Jiazhen Cai, Yu Bao, and **Yeh-Ching Chung**, "What You Can Read is What You Can't Execute," *Computers and Security* (COSE), Vol. 132, July 2023, 103377. (SCI, EI, CCF B)

J6. YongGang Li, Jiazhen Cai, Yu Bao, and **Yeh-Ching Chung**, "What You Can Read is What You Can't Execute," *Computers and Security* (COSE), Vol. 132, July 2023, 103377 (SCI, EI, CCF B)

J7. YuAng Chen and **Yeh-Ching Chung**, "An Unequal Caching Strategy for Shared-Memory Graph Analytics," *IEEE Transaction on Parallel and Distributed Systems* (TPDS), Vol. 34, No. 3, 2023, pp. 955-967. (SCI, EI, CCF A)

J8. YongGang Li, GuoYuan Lin, **Yeh-Ching Chung**, YaoWen Ma, Yi Lu, and, Yu Bao, "MagBox: Keep the Risk Functions Running Safely in a Magic Box," *Future Generation Computer Systems* (FGCS), 140, 2023, pp. 282-298. (SCI, EI, CCF C)

J9. Le-Le Li, Jiang-Yi Liu, Jian-Ping Fan, Xue-Hai Qia. Kai Hwang, **Yeh-Ching Chung**, and Zhi-Bin Yu, "SOCA-DOM: A Mobile SoC Array System for Analyzing Big Data on the Move," *Journal of Computer Science and Technology* (JCST), 37(6): Nov. 2022, pp.1271-1289. (SCI, EI, CCF B)

J10. YuAng Chen and **Yeh-Ching Chung**, "Workload Balancing via Graph Reordering on Multicore Systems," *IEEE Transaction on Parallel and Distributed Systems* (TPDS), Vol. 33, No. 5, 2022, pp. 1231-1245. (SCI, EI, CCF A)

J11. YongGang Li, **Yeh-Ching Chung**, Yu Bao, Yi Lu, ShanQing Guo, and GuoYuan Lin, "KPointer: Keep The Code Pointers On The Stack Point To The Right Code," *Computers and Security* (COSE), Vol. 120, September 2022, 102781. (SCI, EI, CCF B)

J12. Yonggang Li, **Yeh-Ching Chung**, Kai Hwang, and YueJin Li, "Virtual Wall: Filtering Rootkit Attacks To Protect Linux Kernel Functions," *IEEE Transactions on Computers* (TC), Vol. 70, No. 10, 2021, pp. 1640-1653. (SCI, EI, CCF A)

J13. Ching-Hsien Hsu, Kusuma Amilineni, Hao Wu, Christophe Cerin, and **Yeh-Ching Chung**, "A Novel Scalable Network Architecture for the Evolution of Robot Swarm Networks with Dual Connectivity Application in Control-Data Plane," *Swarm and Evolutionary Computation*, Vol. 64, 100889, July 2021. (SCI, EI)

J14. Ching-Hsien Hsu, Xing Chen, Weiwei Lin, Chuntao Jiang, Youhong Zhang, Zhifeng Hao, and **Yeh-Ching Chung**, "Effective Multiple Cancer Disease Diagnosis

Frameworks for Improved Healthcare Using Machine Learning," *Measurement*, Vol. 175, 109145, April 2021. (SCI, EI)

J15. Deverajan Ganesh Gopa, V. Muthukumaran, Ching-Hsien Hsu, Karuppiah Marimuthu, **Yeh-Ching Chung**, and Ying-Huei Chen, "Public Key Encryption with Equality Test for Industrial Internet of Things System in Cloud Computing," *Transactions on Emerging Telecommunications Technologies*, DOI: 10.1002/ett.4202. (SCI, EI)

J16. Tse-Chuan Hsu, Hongji Yang, and **Yeh-Ching Chung**, and Ching-Hsien Hsu, "A Creative IoT Agriculture Platform for Cloud Fog Computing," *Sustainable Computing*: *Informatics and Systems*, Vol. 28, 100285, 2020. (SCI, EI)

J17. Rui Zhang, Wenguang Chen, Tse-Chuan Hsu, Hongji Yang, and **Yeh-Ching Chung,** "ANG – A Combination of Apriori and Graph Computing Techniques for Frequent Itemsets Mining," *The Journal of Supercomputing*, Vol. 75, No. 2, 2019, pp. 646-661 (SCI, EI)

J18. Moise Convolbo, Jerry Chu, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "GEODIS: Towards the Optimization of Data Locality-Aware Jobs Scheduling in Geo-Distributed Data Centers," *Journal of Computing*, Vol. 100, No.1, January 2018, pp. 21-46. (SCI, EI). (SCI, EI)

J19. Fang-Ping and **Yeh-Ching Chung**, "Multi-layer Ontology Based Information Fusion for Situation Awareness," *Applied Intelligence*, Vol. 46, 2017, pp. 285-307. (SCI, EI)

J20. Ying-Ti Liao, Jiazheng Zhou, Chia-Hung Lu, Shih-Chang Chen, Ching-Hsien Hsu, Wenguang Chen, Fong Jiang, and **Yeh-Ching Chung**, "Data Adapter for Uniform Access and Transformation in NoSQL Systems," *Future Generation Computer Systems*, Vol. 65, December 2016, pp. 111-121. (SCI, EI)

J21. Fang-Ping Pai, I-Ching Hsu, and **Yeh-Ching Chung**, "Semantic Web Technology for Agent Interoperability: A Proposed Infrastructure," *Applied Intelligence*, Vol. 44, No. 1, January 2016, pp. 1-16. (SCI, EI)

J22. Che-Lun Hung, Yu-Shiang Lin, Chun-Yuan Lin, **Yeh-Ching Chung**, and Yi-Fang Chung, "CUDA ClustalW: An Efficient Parallel Algorithm for Progressive Multiple Sequence Alignment on Multi-GPUs," *Computational Biology and Chemistry*, Vol. 58, October 2015, pp. 62-68. (SCI, EI)

J23. Yu-Shiang Lin, Chun-Yuan Lin, Che-Lun Hung, **Yeh-Ching Chung**, and Kual-Zheng Lee, "GPU-UPGMA: High-Performance Computing for UPGMA Algorithm Based on Graphics Processing Units," *Concurrency and Computation: Practice and Experience*, Vol. 27, No. 13, September 2015, pp. 3403-3414. (SCI, EI)

J24. Ching-Hsien Hsu, Kenn Slagter, and **Yeh-Ching Chung**, "Locality and Loading Aware Virtual Machine Mapping Techniques for Optimizing Communications in MapReduce Applications," *Future Generation Computer Systems*, Vol. 53, June 2015, pp. 43-54. (SCI, EI)

J25. Kenn Slagter, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "An Adaptive Sampling Mechanism for Partitioning in MapReduce," *International Journal of Parallel Programming*, Vol. 43, No. 3, June, 2015, pp. 489-507. (SCI, EI)

J26. Ssu-Hsuan Lu, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Chig Chung**, "A Scalable P2P Overlay Based on Arrangement Graph with Minimized Overhead," *Peer-to-Peer Networking and Applications*, Vol. 7, No. 4, December 2014, pp. 497-510. (SCI, EI)

J27. Wu-Chun Chung, Hung-Pin Lin, Shih-Chang Chen, Mon-Fong Jang, and **Yeh-Ching Chung**, "JackHare: A Framework for SQL to NoSQL Translation Using MapReduce,"

Automated Software Engineering, Vol. 21, No. 4, December 2014, pp. 489-508. (SCI, EI)

J28. Ching-Jung Hsu, Wu-Chun Chung, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "On Maintenance of Cooperative Overlays in Multi-Overlay Networks," *IET Communications*, Vol. 8, No. 15, October 2014, pp. 2676-2683. (SCI, EI)

J29. Shih-Hsiang Lo, Che-Rung Lee, Quey-Liang Kao, I-Hsin Chung, and **Yeh-Ching Chung**, "Improving GPU Memory Performance with Artificial Barrier Synchronization," *IEEE Transactions on Parallel and Distributed Systems,* Vol. 25, No. 9, September 2014, pp. 2342-2352. (SCI, EI)

J30. Kenn Slagter, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "SmartJoin: A Network-Aware Multiway Join for MapReduce," *Cluster Computing,* Vol. 17. No. 3, September 2014, pp. 629-641. (SCI, EI)

J31. Xuan-Yi Lin and **Yeh-Ching Chung**, "Master-Worker Model for MapReduce Paradigm on the TILE64 Many-Core Platform," *Future Generation Computer Systems*, Vol. 36, July 2014, pp. 19-30. (SCI, EI)

J32. Jiun-Hung Ding, Roger Chien, Shih-Hao Hung, Yi-Lan Lin, Che-Yang Kuo, Ching-Hsien Hsu, and **Yeh-Ching Chung**, "A Framework of Cloud-based Virtual Phones for Secure Intelligent Information Management," *International Journal of Information Management*., Vol. 34, June 2014, pp. 329-335. (SCI, EI)

J33. Ssu-Hsuan Lu, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Chig Chung**, "Effectiveness of a Replica Mechanism to Improve Availability with Arrangement Graph-Based Overlay," *Journal of Network and Computer Applications*, Vol. 41, May 2014, pp. 441-450. (SCI, EI)

J34. Jiun-Hung Ding, Ya-Ting Chang, Zhoudong Kuo, Kuan-Ching Li, and **Yeh-Ching Chung**, "An Efficient and Comprehensive Scheduler on Asymmetric Multicore Architecture Systems," *The Journal of System Architecture*, Vol. 60, No. 3, March 2014, pp. 305-314. (SCI, EI)

J35. Ding-Yong Hong, Jan-Jan Wu, Pen-Chung Yew, Wei-Chung Hsu, Chun-Chen Hsu, Pangfeng Liu, Chien-Min Wang, and **Yeh-Ching Chung**, "Efficient and Retargetable Dynamic Binary Translation on Multicores," *IEEE Transactions on Parallel and Distributed Systems*, Vol. 25, No. 3, March 2014, pp. 622-632. (SCI, EI)

J36. Ching-Hsien Hsu, Kenn Slagter, Shih-Chang Chen, **Yeh-Ching Chung**, "Optimizing Energy Consumption with Task Consolidation in Clouds," *Information Sciences*, Vol. 258, February 2014, pp. 452–462. (SCI, EI)

J37. Po-Chi Shih, Kuo-Chan Huang, Che-Rung Lee, I-Hsin Chung, and **Yeh-Ching Chung,** "TLA: Temporal Look-Ahead Processor Allocation Method for Heterogeneous Multi-Cluster Systems," *Journal of Parallel and Distributed Computing*, Vol. 73, No. 12, December 2013, pp. 1661-1672. (SCI, EI)

J38. Kenn Slagter, Ching-Hsien Hsu, and **Yeh-Ching Chung**, Daqiang Zhang, "An Improved Partitioning Mechanism for Optimizing Massive Data Analysis Using MapReduce," *The Journal of Supercomputing*, Vol. 66, No. 1, October 2013, pp. 539-555. (SCI, EI)

J39. Wu-Chung Chung, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "Direction-Aware Resource Discovery in Large-Scale Distributed Computing Environments," *The Journal of Supercomputing*, Vol. 66, No. 1, October 2013, pp. 229-248. (SCI, EI)

J40. Yu-Wei Chan, Feng-Tsun Chien, Ronald Y. Chang, Min-Kuan Chang, **Yeh-Ching Chung**, "Spectrum Sharing in Multi-Channel Cooperative Cognitive Radio Networks: A Coalitional Game Approach," *ACM/Springer Wireless Networks*, Vol. 19, No. 7, October 2013, pp. 1553-1562. (SCI, EI)

J41. Xuan-Yi Lin, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "Efficient Programming Paradigm for Video Streaming Processing on TILE64 Platform," *The Journal of Supercomputing*, Vol. 65, No. 2, August 2013, pp. 823-847. (SCI, EI)

J42. Shih-Hsiang Lo, Che-Rung Lee, I-Hsin Chung, and **Yeh-Ching Chung**, "Optimizing Pairwise Box Intersection Checking on GPUs for Large-Scale Simulations," *ACM Transactions on Modeling and Computer Simulation*, Vol. 23, No. 3, Article 19, July 2013. (SCI, EI)

J43. Wu-Chun Chung, Yi-Hsiang Lin, Kuan-Chou Lai, Kuan-Ching Li, and **Yeh-Ching Chung**, "SARIDS: A Self-Adaptive Resource Index and Discovery System," *International Journal of Ad Hoc and Ubiquitous Computing*, Vo. 10, No. 2, July 2012, pp. 74-83(10). (SCI, EI)

J44. Po-Chi Shih, Kuo-Chan Huang, and **Yeh-Ching Chung**, "Improving Processor Allocation in Heterogeneous Computing Grid through Considering Both Speed Heterogeneity and Resource Fragmentation," *The Journal of Supercomputing*, Vol. 60, No. 3, May 2012, pp. 311-337. (SCI, EI)

J45. Jiazheng Zhou, Shen-En Liu, and **Yeh-Ching Chung**, "A Hardware Supported Multicast Scheme Based on XY Routing for 2-D Mesh InfiniBand Networks," *The Journal of Supercomputing*, Vol. 59, No. 2, February 2012, pp. 901-919. (SCI, EI)

J46. Jiazheng Zhou and **Yeh-Ching Chung**, "Tree-Turn Routing: An Efficient Deadlock-Free Routing Algorithm for Irregular Networks," *The Journal of Supercomputing*, Vol. 59, No. 2, February 2012, pp. 882-900. (SCI, EI)

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C137. **Yeh-Ching Chung**, Yaa-Jyun Yeh, and J.-S.Liu, "A Parallel Dynamic Load Balancing Algorithm for Solution-Adaptive Finite Element Meshes on 2D-Tori," *Proceedings of International Computer Symposiums* (ICS), Taiwan, Dec. 1994, pp.967-972. (NSC84-408-E-035-005)

C138. **Yeh-Ching Chung**, Yaa-Jyun Yeh, and Chia-Cheng Liu, "A Parallel Run-Time Iterative Load Balancing Algorithms for Solution-Adaptive Finite Element Meshes on Hypercubes," *Proceedings of IEEE International Conference on Parallel and Distributed Systems* (ICPADS), Taiwan, Dec. 1994, pp.210-215. (NSC84-408-E-035-005)

C139. **Yeh-Ching Chung** and Sanjay Ranka, "A Compile-Time Optimization Approach for

Scheduling of Tasks on Distributed Memory Multiprocessors," *Proceedings of IEEE International Conference on Parallel and Distributed Systems* (ICPADS), Taiwan, Dec. 1993, pp.87-91.

C140. **Yeh-Ching Chung** and Sanjay Ranka, "A Parallel Program Design Tool for Single Program Multiple Data Model on Distributed Memory Multiprocessors," *Proceedings of IEEE Hawaii International Conference on System Sciences*, USA, Jan. 1993, pp.433-434.

C141. **Yeh-Ching Chung** and Sanjay Ranka, "Applications and Performance Analysis of A Compile-Time Optimization Approach for List Scheduling Algorithms on Distributed Memory Multiprocessors," *Proceedings of IEEE Supercomputing*, USA, Nov. 1992, pp. 512-521.

C142. Roger Chen and **Yeh-Ching Chung**, "Embedding Networks with Ring Connections in Hypercube Machine," *Proceedings of IEEE International Conference on Parallel Processing* (ICPP), Vol. III, USA, August 1990, pp. 327-334.

C143. **Yeh-Ching Chung** and Sanjay Ranka, "Mapping Finite Element Graphs onto Hypercubes," *Proceedings of IEEE Symposium on the Frontiers of Massively Parallel Computation*, USA, May 1990, pp. 135-144.

Supervised Ph.D. Students

- Ching-Hsien Hsu, January 1999
  Dissertation Title – Efficient Methods for Run-Time Array Redistribution
- Ching-Jung Liao, June 1999
  Dissertation Title – Efficient Partitioning and Load Balancing Methods for Finite Element Graphs on Distributed Memory Multicomputers
- Chun-Yuan Lin, June 2003
  Dissertation Title – Efficient Data Parallel Program Design for Multi-Dimensional Array Operations Based on the Extended Karnaugh Map Representation Scheme
- Chin-Feng Lin, January 2004
  Dissertation Title – Efficient Parallel Volume Rendering Methods on Distributed Memory Multicomputers
- Shih-Kuan Liao, June 2004
  Dissertation Title – Time Varying Volume Data Rendering
- Che-Lun Hung, March 2010
  Dissertation Title – Efficient Algorithms for Triple-wise Alignment and its Applications
- Chun-Hsien Wu, June 2010
  Dissertation Title – Deployment Strategies for Wireless Sensor Networks
- Jia-Zheng Zhou, March 2011
  Dissertation Title – Routing in Interconnection Networks and Its Applications
- Po-Chi Shih, July 2012
  Dissertation Title – Improving Parallel Job Scheduling Performance in Multi-Cluster through Look-Ahead Processor Allocation

- Shih-Hsiang Lo, March 2013
  Dissertation Title – Optimization Techniques for Reducing Data Communication Costs on GPU Systems
- Wu-Chun Chung, April 2013
  Dissertation Title – Decentralized Management of Resource Information and Discovery in Large-Scale Distributed Computing
- Ding-Yong Hong, April 2013
  Dissertation Title – Efficient and Retargetable Dynamic Binary Translation
- Xuan-Yi Lin, June 2013
  Dissertation Title – Master-Worker Paradigm on TILE64 Many-Core Platform
- Yu-Wei Chan, June 2013
  Dissertation Title – Resource Sharing Strategies for Wide-Area Fusing Locality-Aware and Game Theoretic Approaches
- Kenn Slagter, November 2013
  Dissertation Title – Load Balancing and Data Placement in MapReduce
- Ssu-Hsuan Lu, January 2014
  Dissertation Title – A Self-Scaling Overlay Network Based on Arrangement Graph
- Fang-Ping Bai, July 2016
  Dissertation Title – An Ontology-Driven Multi-Agent Information Fusion Framework to Enhance Situation Awareness
- Moïse Wendkuuni Convolbo, July 2017
  Dissertation Title – Scheduling Techniques on Cloud Resources under Cost and Performance Requirements
- Yu-Hsiang Lin, January 2018
  Dissertation Title – A New API Remoting Policy for General-Purpose Computing on GPU Virtualization
- YuAng Chen, May 2022
  Dissertation Title – High-Performance Graph Processing on Multicore Systems
- Ao Shan, November 2023
  Dissertation Title – Imbalanced Data Classification Problems Based on Data Complexity
- Weibin Chen, September 2024
  Dissertation Title – Profile-Guided Optimization for Function Reordering: Algorithm and Application
- Lele Li, December 2024
  Dissertation Title – On the Co-optimization of Performance and Resource Efficiency for Big Data Analytics
- Renwen Ma, January 2025 (Co-Supervised with Professor Kai Hwang)
  Dissertation Title – Securing Federated Machine Learning Systems with Blockchain and Zero-Knowledge Proofs
- Shihao Hong, May 2025
  Dissertation Title – Hardware-Aware Redundancy Pruning for DNN Inference: From ReRAM PIM to GPU

Supervised Master Students

- Yaa-Jyun Yeh, June 1994
  Thesis Title – Parallel Dynamic Load Balancing Algorithms for Solution-Adaptive Finite Element Graphs on Distributed Memory Multiprocessors
- Wu-Hsun Ho, June 1995
  Thesis Title – PPT: A Parallel Programming Tool for Distributed Memory Multiprocessors
- Shih-Jey Chen, June 1996
  Thesis Title – HCL: An Implementation of Message Passing Interface on Workstation Clusters
- Sheng-Wen Bai, June 1998
  Thesis Title – A Generalized Basic-Cycle Calculation Technique for Efficient Array Redistribution
- Jen-Chih Yu, June 1999
  Thesis Title – Efficient Compositing Methods for the Sort-Last-Sparse Parallel Volume Rendering System on Distributed Memory Multicomputers
- Chih-Chang Chen, June 1999
  Thesis Title – A Dynamic Diffusion Optimization Method for Irregular Finite Element Graph Partitioning
- Chao-Jen Lee, June 2000
  Thesis Title – A Web-Based Parallel PDE Solver Generation System for Distributed Memory Computing Environments
- Chun-Yuan Lin, June 2000
  Thesis Title – Efficient Algorithms for Multi-Dimensional Matrix Operations
- Ching-Ting Pan, June 2001
  Thesis Title – An Efficient Hash Based Method for Discovering the Maximal Frequent Set
- Su-Ming Yang, June 2001
  Thesis Title – The Design and Implementation of an IAM2000s C Compiler
- Wu-Hong Chen, June 2001
  Thesis Title – A Real-Time Flight Control Embedded System Based on uC/OS-II Kernel
- Ching-Jung Long, January 2002
  Thesis Title – Efficient Load Balancing Methods for Cellular Mobile Networks
- Chun-Hsing Chang, June 2002
  Thesis Title – A Configurable Hard Real-Time Operating System Based on uC/OS-II Kernel for Flight Controls
- Yuan-Chang Tsai, June 2002
  Thesis Title – An Integrated Design for Automatic System Development Tool Kits Generation
- Chun-Hsien Wu, June 2002
  Thesis Title – A Service Composition Framework for Service Discovery/Delivery Based on SLP
- Kuen-Lin Cheng, June 2002
  Thesis Title – The Integration of GPS and GIS on PDAs

- Jen-Shen Lin, June 2002
  Thesis Title – Image Compression by Adjacent Decomposed Bit-Plane Substitution
- Xuan-Yi Lin, June 2003
  Thesis Title – An Efficient Communication Scheme for Fat-Tree Topology on InfiniBand Networks
- Yao-Ming Sun, July 2004
  Thesis Title – An Efficient Deadlock-Free Tree-Based Routing Algorithm for Irregular Wormhole-Routed Networks Based on the Turn Model
- Jia-Zheng Zhou, July 2004
  Thesis Title – Multicast in Fat-Tree-Based InfiniBand Networks
- Chang-Yi Lin, July 2004
  Thesis Title – An Adaptive Migratory Home Protocol for Software Distributed Shared Memory Systems
- Yun-Chan Lee, July 2005
  Thesis Title – A Hybrid Wavelet-Based Video Codec with Low Computational Complexity, Memory Efficiency, and Scalability
- Ssu-Chun Tseng, July 2005
  Thesis Title – A Scalable and Robust Scheme Based on the Multiple Description Coding for Low Bit-Rate Speech Coders
- Tse-Ming Kao, July 2005
  Thesis Title – A Multithreaded DSM System with Thread Migration for Memory Locality over InfiniBand
- Kuo-Chuan Lee, July 2005
  Thesis Title – A Delaunay Triangulation Based Method for Wireless Sensor Network Deployment
- Hsien-Ting Kuo, July 2005
  Thesis Title – An Efficient and Low Overhead Random Forwarding Table Construction Method for Deadlock-Free Routing Algorithms in InfiniBand Networks
- Ding-Yong Hong, July 2005
  Thesis Title – An Efficient MPI-IO for Noncontiguous I/O over InfiniBand
- Shen-En Liu, July 2005
  Thesis Title – Hardware Supported Multicast in Tow-Dimensional Mesh InfiniBand Networks
- Wei-Ching Kao, September 2005
  Thesis Title – An embedded Print Server
- Yang-Ping Cheng, September 2005
  Thesis Title – A Performance-Efficient Task Duplication- Based Scheduling Algorithm for Heterogeneous Computing
- Tse-Yu Hsu, June 2006
  Thesis Title – A Register Allocation Algorithm for Shared and Clustered Hybrid Register File Organization
- Meng-Ru Lin, June 2006
  Thesis Title – CFR: A Peer-to-Peer Collaborative File Repository System
- Ching-Wen Yu, June 2006

Thesis Title – SPE: A Programming Environment for Wireless Sensor Networks

- Yi-Kai Pong, June 2006
  Thesis Title – Applying Curve Fitting Techniques to Construct the Synopsis of Data Streams
- Jiun-Hung Ding, June 2006
  Thesis Title – SEMU: A Framework of Simulation Environment for Wireless Sensor Networks
- Shih-Hsiang Lo, June 2006
  Thesis Title – Cooperative Localization with Pre-Knowledge using Bayesian Network for Wireless Sensor Networks
- Wen-Li Tang, June 2006
  Thesis Title – A Fast, Flexible, and Precise Simulation Framework for Starfish DSP Architecture
- Chih-Han Lai, June 2007
  Thesis Title – A Construction of Peer-to-Peer Streaming System Based on Flexible Locality-Aware Overlay Networks
- Tun-Yi Chang, June 2007
  Thesis Title – Taking Advantage of Multi-Core Architecture in Disk Array
- Yu-Chen Wu, July 2007
  Thesis Title – Dynamic Helper Threaded Prefetching on the Multi-Core Software RAID
- Chin-Wei Tang, July 2007
  Thesis Title – A System-Level Simulation Framework for Performance Evaluation of Hardware-Based Network Intrusion Detection Systems
- Meng-Hsuan Cheng, July 2007
  Thesis Title – A VLIW-Based Post Compilation Framework for Multimedia Embedded DSPs with Hardware Specific Optimizations
- Wei-Lun Tsai, July 2007
  Thesis Title – A Tag-Based Overlay Network for Defending Against DDoS Attack
- Yan-Zu Wu, July 2007
  Thesis Title – A Extended Shared and Clustered Register Files Organization – Wired SCRF Architecture
- Chung Han Lin, July 2007
  Thesis Title – M-Ring: A Distributed, Self-Organize, Load Balancing Communication Method on Super Peer Network
- Sheng-Je Hung, July 2007
  Thesis Title – A Pipeline and Macroblock-Level Parallelism Architecture Design for H.264/AVC Baseline Decoder
- Yi-Hsin Lin, July 2008
  Thesis Title – Reduce Leakage Power of JPEG Image on Asymmetric SRAM
- Chung-Hsuan Ho, July 2008
  Thesis Title – Malugo: A Peer-to-Peer Storage System
- Meng-Ju Hsieh, July 2008
  Thesis Title – Minimize the Distance of Failure Location for Automatic Debugging Technique with Improved State-Coverage mechanism

- Chih-Chen Chen, July 2008
  Thesis Title – Using parity cache to reduce overhead of small write requests under Linux software RAID-5 system
- Hao Yuan, July 2008
  Thesis Title – Profile Based Trace Selection of Code Layout Optimizations in Open64 Compiler
- Cheng-Tin Tsai, July 2008
  Thesis Title – A Parallel Algorithm of H.264/AVC Encoder on Multi-Core Systems
- Cheng-An Chiu, July 2008
  Thesis Title – An Adaptive Method for Distributed Data Management in High Level Architecture
- Fu-Mao Chen, July 2008
  Thesis Title – A Peer-to-Peer High Level Architecture
- Chih-Chien Chang, June 2009
  Thesis Title – Helper Threaded Shadow Paging on Multi-Core Virtualization Processors
- Chin-Jung Hsu, June 2009
  Thesis Title – Cooperative Maintenance in Multi-Overlay Environments
- Yi-Hsiang Lin, June 2009
  Thesis Title – SARIDS：A Self-Adaptive Resource Index and Discovery System
- Ze-Chuan Yeh, June 2009
  Thesis Title – A Power-Efficient and High-Throughput Deblocking Filter Hardware Architecture for H.264/AVC
- Chia-Dar Tsai, July 2009
  Thesis Title – A Low-Power Hardware Design for H.264/AVC Baseline Decoder
- Chih-Yin Wu, July 2009
  Thesis Title – Dryad: An Ahead of Time Compiler Optimization Framework for Android
- Min-Chuan Hung, July 2009
  Thesis Title – A Time/Space Balanced Flash Translation Layer for Embedded Systems
- Shuo-Hung Chen, July 2010
  Thesis Title – HW/SW Co-Design of High-Performance Multi-Core Vector Graphics System
- Wei-Chao Chang, July 2010
  Thesis Title – A Locality-based Multicast Scheme on Structured Peer-to-peer Overlay for High Level Architecture
- Chia-Hong Chien, September 2010
  Thesis Title – A Separate Code Cache Model for a Parallel Multi-Core System Emulator Based on QEMU
- Hsiang-Wei Sung, September 2010
  Thesis Title – Processor virtualization and optimization for KVM-ARM
- Fan-Lei Liao, July 2011
  Thesis Title – Memory Virtualization Design for KVM-ARM
- Tai-Wen Lung, July 2011
  Thesis Title – Data Locality Optimizations for Multi-Level Caches in Java Multi-Core Compiler

- Chih-Sheng Wang, July 2011
  Thesis Title – An Ahead-of-Time compilation and optimization framework based on LLVM for Google Android
- Moise Convolbo, July 2011
  Thesis Title – Rsvg: Display Big Size Vector Map on Mobile Device
- Ya-Ting Chang, May 2012
  Thesis Title – NUMA-Aware Scheduler for Asymmetric Multicore Processors
- Guillermo A. Perez, May 2012
  Thesis Title – A Hybrid Just-In-Time Compiler for Android
- Ming-Hsuan Lee, June 2012
  Thesis Title – The Scheduling and Timer Interrupt Mechanism for PQEMU
- Chao-Chu Wang, June 2012
  Thesis Title – Implementation of x86 Atomic Instruction Emulation in PQEMU
- Jia-Wang Chung, June 2012
  Thesis Title – Interrup Controller Design Based on PQEMU
- Chang-Jung Lin, June 2012
  Thesis Title – I/O Performance Optimization and Analysis for ARMvisor Using Virtio and Irqchip
- Yu-Ying Chen, June 2012
  Thesis Title – Improve 3D Performance of Qemu-Spice
- Shih-Wei Li, June 2012
  Thesis Title – Memory Virtualization for ARMvisor
- Hung-Ping Lin, June 2012
  Thesis Title – Structured Data Processing on MapReduce in NoSQL Database
- Yi-Man Ma, July 2012
  Thesis Title – InfiniBand Virtualization for Kernel-based Virtual Machine
- Zhoudong Guo, June 2013
  Thesis Title – HSA Emulator Design Based on QEMU
- Sheriffo Ceesay, June 2013
  Thesis Title – Moving Towards Pure ANSI SQL in NoSQL
- Chih-Hao Chang, July 2013
  Thesis Title – Memory Virtualization on ARM Architecture with Virtualization Hardware Extension
- Chung-Min Kao, July 2013
  Thesis Title – The LLVM Based GPU Compiler in Heterogeneous System Architecture Emulator: HTranslator
- Ping-Hao Chang, July 2013
  Thesis Title – CPU Virtualization with ARM Virtualization Extension on ARMvsior
- Shu-Ming Chang, July 2014
  Thesis Title – Adaptive Fusion SQL Engine on Hadoop Framework

- Ying-Ti Liao, July 2014
  Thesis Title – Data Adapter for Querying and Transformation Between SQL and NoSQL Databases
- Su-Shien Ho, July 2014
  Thesis Title – Distributed Metaserver Mechanism and Recovery Mechanism Support in Quanteast File System
- Yi-chen Yang, July 2014
  Thesis Title – CSCJ: Star Join Processing on HBase Using Hadoop MapReduce
- Intan Maghfirah, July 2014
  Thesis Title – Constraints Preserving in Schema Transformation to Enhance Database Migration from MySQL to HBase
- Ngo-Thi Nha, July 2014
  Thesis Title – MTSH: Multi-Tenant Web-based Management System on SSDB and HDFS
- Kuo-Min Lin, July 2014
  Thesis Title – A Compilation Framework for HSA
- Bai-Cheng Jeng, July 2014
  Thesis Title – HSAemu Framework
- Che-Yang Kuo, July 2015
  Thesis Title – Implementation of Image Feature Supports in HSAemu Framework
- Jui Hsiao, July 2015
  Thesis Title – An OpenCL 2.0 Compilation Framework for HSA
- Fei Peng, July 2015
  Thesis Title – OpenCL 2.0 Enabled HSA Hardware Emulation
- Yen-Yu Lin, July 2015
  Thesis Title – Container-Based Automated Workflow System Supporting On-Demand Service Provision
- Hsing-Chang Chou, July 2015
  Thesis Title – Container-Based Scale-Out Architecture for Cloud Storage Service
- Wei-Chih Sun, July 2015
  Thesis Title – An OpenCL 2.0 Runtime Based on HSA Runtime
- Lyu-Wei Wang, July 2015
  Thesis Title – BiFennel: Fast Graph Partitioning Algorithm for Big Data
- Chih-Hsien Chen, July 2015
  Thesis Title – SQL/PCM Design and Implement on NoSQL Database
- Yi-Lan Lin, July 2015
  Thesis Title – Community-Based M2M Framework Using Smart/HetNet GW
- Chun-Feng Wu, June 2016
  Thesis Title – A Hybrid Mechanism to Improve Write Scenarios for Cloud Storage Services

- Mu-Han Huang, June 2016
  Thesis Title – Application-Aware Resource Provisioning for On-Demand Container-Based Cloud Radio Access Network
- Chia-Hung Lu, June 2016
  Thesis Title – Log-Based Synchronization Mechanism Between RDBMS and NoSQL Databases
- Luis Herrera, June 2016
  Thesis Title – A KVM-Based GPGPU Virtualization Technique for Windows
- Hao-Che Hsu, June 2016
  Thesis Title – An HSA Emulation Platform with SVM and Multiple Computing Devices Support
- Beibei Li, June 2016
  Thesis Title – Container-Based Workflow System for Production Server Deployment
- Shih-Chun Huang, July 2017
  Thesis Title – Application-Aware Offloading Mechanism for Mobile Edge Computing
- Yu-Chin Luo, July 2017
  Thesis Title – Improve Resource Utilization of Cloud Radio Access Network Using Containerization and Functional Split Techniques
- Bing-Liang Chen, July 2017
  Thesis Title – Dynamic Module Deployment Mechanism and Placement Optimization on M2M Platform
- Hung-Ying Tai, July 2017
  Thesis Title – Crystal: An Extensible Framework for Profiling and Optimizing Large-Scale System Software
- Meng-Hsun Yang, July 2017
  Thesis Title – Code Layout Optimization Applying Community Detection Algorithm
- Jian-Lin Chen, July 2017
  Thesis Title – Function Reordering Optimization Based on Weighted Call Graph for Large-Scale Software in LLVM
- Wenyu Huang, July 2023
  Thesis Title – ReKV: A Good Read Performance LSM-TreeKey-Value Store
- Yufan Yang, December 2023
  Thesis Title – HWO: High-Order Based and WCO Guaranteed Query Plan Optimizer for Graph Pattern Matching
- Yihan Liu, December 2024
  Thesis Title – V2M: Automatic Music Generation for a Given Video Based on Video Understanding and Emotion Detection using a Multimodal Transformer Model

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Keynote/Tutorial Speech (since 2010)

- Keynote Speaker, IEEE Big Data Congress Taipei Satellite Session, Taipei, Taiwan, 2015

Title – SSBDS: A High-Performance Big Data Service for Research and Education

- HSA Tutorial Speaker, International Symposiums on Computer Architecture (ISCA), Minneapolis, USA, 2014

  Title – HSA Runtime

- Keynote Speaker, China Conference on Cloud Computing, Guangzhou, China, 2013

  Title – UniCloud: A Community/Hybrid Cloud Platform for Research and Education

- Keynote Speaker, International Conference on Computational Intelligence, Modeling and Simulation, Bali, Indonesia, 2010

  Title – Computer Go Game

Conference Chairmanships (Since 2006)

- Advisory Committee Member of International Conference on Big Data Intelligence and Computing (DataCom) 2019, November 18-21, 2019
- General Co-Chair of International Conference on Big Data Intelligence and Computing (DataCom) 2015, December 19-21, 2015
- Program Co-Chair of IEEE International Conference on Cloud Computing and Big Data (CCBD), December 4 – 6, 2015
- Program Co-Chair of IEEE International Conference on Parallel and Distributed Systems (ICPADS), December 16 – 19, 2014
- General Co-Chair of IEEE International Conference on Cloud Computing Technology and Science (CouldCom), December 3-6, 2012
- Track Co-Chair of International Conference on Parallel Processing (ICPP), September 13-16, 2011
- Track Co-Chair of International Conference on Contemporary Computing (IC3), August 8-10, 2010
- General Co-Chair of International Symposium on Information Assurance and Security Management (IASM), December 14-16, 2009
- Track Co-Chair of International Symposium on Pervasive Systems, Algorithms, and Networks (I-SPAN), December 14-16, 2009
- Workshop Co-Chair of National Computer Symposium (NCS) 2009, November 27-28, 2009
- Track Co-Chair of International Conference on Contemporary Computing (IC3), August 17-19, 2009
- Program Co-Chair of International Symposium on Grid and Distributed Computing (GDC), December 13-15, 2008
- Award Chair of IEEE Asia-Pacific Services Computing Conference (APSCC), December 09-12, 2008
- Program Co-Chair of IEEE Asia-Pacific Computer Systems Architecture Conference (ACSAC), August 20-22, 2008
- Program Co-Chair of International Conference on Grid and Pervasive Computing (GPC

2006), May 2-4, 2006

Program Committee memberships (Since 2007)

- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, May 28 - June 1, 2023
- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, May 16-11, 2022
- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, June 14-18, 2021
- IEEE International Conference on Communications (ICC-2020) SAC Symposium on Big Data Track, June 7-11, 2020
- IEEE International Conference on Communications (ICC-2018) SAC Symposium on Big Data Track, May 20-24, 2018
- IEEE International Global Communications Conference (Globecom-2017) SAC on Big Data Track, December 4-8, 2017
- IEEE International Conference on Communications (ICC-2017) SAC Symposium on Big Data Networking Track, May 21-25, 2017
- IEEE International Global Communications Conference (Globecom-2016) SAC on Big Data Track, December 4-8, 2016
- International Conference on Network and Parallel Computing (NPC-2016), October 28-29, 2016
- International Conference on Green and Pervasive, and Cloud Computing (GPC-2015), May 4-6, 2015
- IEEE International Conference on Cloud Engineering (IC2E-2015), March 9-12, 2015
- IEEE International Conference on Green Computing and Communications (GreenCom-2014), September 1-3, 2014
- IEEE International Conference on Embedded Computing (EmbeddedCom-2014), August 24-27, 2014
- International Conference on Green and Pervasive, and Cloud Computing (GPC-2014), May 26-28, 2014
- IEEE International Conference on Green Computing and Communications (GreenCom-2013), August 20-23, 2013
- International Conference on Smart Grids and Green IT Systems (SMARTGREENS-2013), May 9-10, 2013
- International Conference on Grid and Pervasive computing (GPC-2013), May 9-11, 2013
- IEEE Asia Pacific Cloud Computing Congress (APCloud-2012), November 14-17, 2012
- IEEE International Parallel & Distributed Processing Symposium (IPDPS-2012) May 21-25, 2012
- International Conference on Grid and Pervasive Computing (GPC-2012), May 11-13, 2012
- IEEE International Conference on High Performance Computing (HiPC-2011), December 19-22, 2011

- International Symposium on Grid and Distributed Computing (GDC-2011), December 8-10, 2011
- International Conference on Network and Parallel Computing (NPC-2011), October 21-23, 2011
- International Conference on Contemporary Computing (IC3-2011), August 8-10, 2011
- IEEE/ACM International Conference on Green Computing and Communications (GreenCom-2011), August 4-5, 2010.
- IEEE International Symposium on Parallel and Distributed Processing with Applications (ISPA-2011), May 26-28, 2011.
- International Conference on Grid and Pervasive Computing (GPC-2011), May 11-13, 2011
- International Symposium on Grid and Distributed Computing (GDC-2010), November 11-13, 2010
- IEEE/ACM International Conference on Green Computing and Communications (GreenCom-2010), October 30-November 1, 2010.
- International Conference on Advances in P2P Systems (AP2PS-2010), October 25 - 30, 2010.
- IEEE International Conference on Embedded Software and Systems (ICESS-2010) June 29-July 1, 2010.
- International Conference on Grid and Pervasive Computing (GPC-2010), May 10-13, 2010
- International Conference on Frontier of Computer Science and Technology (FCST-2009), December 17-19, 2009
- IEEE International Conference on High Performance Computing (HiPC-2009), December 16-19, 2009
- International Symposium on Grid and Distributed Computing (GDC-2009), December 10-12, 2009
- International Conference on Embedded and Multimedia Computing (EM-Com-2009), December 10 - 12, 2009
- International Conference on Computer Science and its Applications (CSA-2009), December 10 - 12, 2009
- International Conference on Parallel and Distributed Systems (ICPADS-2009), December 9 – 11, 2009
- International Conference on Network and Parallel Computing (NPC-2009), October 19-21, 2009
- International Conference on Advances in Peer-to-Peer Systems (AP2PS-202009), October 11-16, 2009
- International Conference on Embedded Software and Systems (ICESS-2009), May 25-27, 2009
- International Workshop on Service-Oriented P2P Networks (ServP2P-2009), May 18-21, 2009
- International Conference on Grid and Pervasive Computing (GPC-2009), May 4-8, 2009
- International Conference on Information Systems, Technology and Management

(ICISTM-2009), March 12-13, 2009

- High Performance Computing Asia 2009, March 2-6, 2008
- IEEE International Conference on High Performance Computing (HiPC-2008), December 17-20, 2008
- International Conference on Embedded and Ubiquitous Computing Conference (EUC-2008), December 17-20, 2008
- International Symposium on Ubiquitous Multimedia Computing (UMC-2008), October 14-16, 2008
- IEEE International Symposium on Embedded Computing (SEC-2008), October 6-8, 2008
- International Conference on Network and Parallel Computing (NPC-2008), September 18-21, 2008
- International Conference on Contemporary Computing (IC3-2008), August 7-9, 2008
- International Conference on Embedded Software and Systems (ICESS-2008), July 28-31, 2008
- International Conference on Computational Science and Engineering (CSE-2008), June 25-27, 2008
- International Conference on Communications, Circuits, and Systems (ICCCAS-2008), May 25-27, 2008
- International Conference on Grid and Pervasive Computing (GPC-2008), May 25-28, 2008
- International Conference on Advanced Information Networking and Applications (AINA-2008), March 25-28, 2008
- International Conference on Information Systems, Technology and Management (ICISTM-2008), March 6-8, 2008
- International Conference on Embedded and Ubiquitous Computing Conference (EUC-2007), December 10-13, 2007
- International Conference on Grid and Pervasive Computing (GPC-2007), May 2-4, 2007

University and Departmental Service at NTHU

- Deputy director of library, December 2003 – July 2012
- Deputy director of computer and communication research center, August 2008 – July 2015
- Ph.D. admission committee member, August 2002 – July 2015
- Master admission committee member, August 2002 – July 2015
- Faculty recruiting committee member, August 2002 – July 2009

University and Department Service at FCU

- Chairman of Department of Information Engineering and Computer Science, August 1998 – July 2001
- Network Division Director of Computer Center, January 1994 – July 1997

- Education Division Director of Computer Center, August 1993 – December 1993
- PhD admission committee member, August 1994 – July 2002
- Master admission committee member, August 1992 – July 2002

Teaching Activities

Courses taught at CUHKSZ

- CSC3050 – Computer Architecture,
- CSC3150 – Operating Systems,
- CSC4005 – Distributed and Parallel Computing,
- CSC4180 – Compiler Construction
- CSC6230 – Advanced Operating Systems,

Courses taught at NTHU

- CS135502 – Introduction to C Programming Language,
- CS235102 – Data Structure,
- CS242202 – Assembly Language and System Programming,
- CS340402 – Compiler Construction,
- CS342302 – Operating Systems.
- CS510400 – Embedded Software for Networked SoC Systems,
- CS510500 – Embedded Processor Design,
- CS541000 – Virtualization Techniques,
- CS542100 – Cloud Computing,
- CS542200 – Parallel Programming,
- CS542800 – Heterogeneous System Architecture.
